

Risk Register & Closure Note

B.Tech CSE 2025–29 • Software Engineering & Project Management • Semester III

Dashboard basis: The project author has developed three dashboards—Student, Faculty and Admin. This document uses the supplied walkthrough to describe them and distinguishes those dashboard details from requirements, integrations or metrics that are proposed or not independently verified.

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1. Scoring method

Probability and impact use a 1–5 scale (1 = rare/very low, 5 = very likely/very high). **Exposure = probability × impact**, maximum 25. Scores are planning judgements based on the supplied scenario, not measured statistics. Review them with stakeholders at each milestone.

2. Exposure-ranked risk register

Rank	ID / risk	P	I	Exposure
1	R1 — Faculty recording availability is below the schedule assumption; only 8 courses are stated as recordable in the next 3 months.	4	5	20
2	R2 — Editing capacity is interrupted or actual rework exceeds the 60 hours/week assumption.	3	5	15
3	R3 — Payment status is incorrectly trusted or provider callback integration fails.	3	5	15
4	R6 — Firestore and local JSON dual-write paths become inconsistent or lose updates.	3	4	12
5	R9 — NFR targets or privacy/retention rules are not approved before production.	3	4	12

Rank	ID / risk	P	I	Exposure
6	R4 — Student records, credentials or role-protected data are exposed.	2	5	10
7	R7 — Certificate eligibility logic issues an award to a learner who has not met both rules.	2	5	10
8	R8 — Revenue targets are mistaken for guaranteed collections.	3	3	9
9	R5 — Video hosting/CDN is unavailable or performs poorly for learners.	2	4	8

RISK RESPONSES AND OWNERS

Risk	Response / mitigation	Owner	Trigger / contingency
R1 — Faculty recording availability is below the schedule assumption; only 8 courses are stated as recordable in the next 3 months.	Mitigate: reserve dated recording slots, name a backup faculty contact, approve course order, and review actual sessions weekly. Avoid scope promise: do not announce all 30 at W10.	Faculty coordinator	If fewer than 8 course packages remain on track for W10, retain phased scope and move affected courses to the next release.
R2 — Editing capacity is interrupted or actual rework exceeds the 60 hours/week assumption.	Mitigate: track editor hours and first-pass approval rate; standardize templates and quality checks; protect editor capacity from unrelated work.	Production lead	If actual throughput falls below 60 effort-hours/week for two consecutive weeks, reforecast the W18 catalogue date.
R3 — Payment status is incorrectly trusted or provider callback integration fails.	Mitigate: validate signed provider callbacks server-side, use idempotency keys, reconcile transactions daily and test failure/timeout paths.	Payment/IT owner	Any mismatch between provider settlement and platform status: pause affected enrolments, reconcile and notify finance.
R6 — Firestore and local JSON dual-write paths become inconsistent or lose updates.	Mitigate: select an authoritative store, make writes idempotent, record reconciliation events, back up daily and run restore drills.	Backend / IT lead	Reconciliation mismatch or failed restore: pause dependent writes and recover from the authoritative copy.
R9 — NFR targets or privacy/retention rules are not approved before production.	Mitigate: obtain sign-off from IT, privacy and academic owners; retain open decisions in the release checklist.	Project lead	Any unresolved Must-level NFR or data policy at release gate: defer production launch or obtain formal decision.
R4 — Student records, credentials or role-protected data are exposed.	Mitigate: least privilege, adaptive password hashing, TLS, secret management, access tests and audit logging; rehearse incident response.	Security / IT lead	Any unauthorised access alert: contain access, preserve logs and follow university incident policy.
R7 — Certificate eligibility logic issues an award to a learner who has not met both rules.	Mitigate: enforce eligibility server-side, add decision-table and boundary tests, use idempotent issuance and audit each certificate record.	QA lead + academic owner	Any invalid certificate: suspend issuance, identify affected records and follow university revocation/correction policy.
R8 — Revenue targets are mistaken for guaranteed collections.	Mitigate: label dashboard figures as projections; reconcile actual paid transactions and report enrolment assumptions separately.	Director / finance owner	Actual enrolments diverge from target: revise forecast without changing the case's stated inputs.

Risk	Response / mitigation	Owner	Trigger / contingency
R5 — Video hosting/CDN is unavailable or performs poorly for learners.	Mitigate: select an accountable CDN, measure start time and availability, monitor playback errors, and keep re-upload/rollback instructions.	IT operations	Availability below 99.5% monthly or p95 start time above 3 seconds: investigate provider and communicate service status.

3. RMMM plans for the top three risks

R1 — Faculty availability

- **Risk reduction:** build a named course-to-faculty schedule; confirm dates and backup participants; prioritize the first 8 courses by demand and readiness; record attendance and content approval weekly.
- **Monitoring:** planned versus completed recording sessions, approved hours, and number of courses ready for edit/review.
- **Management / contingency:** keep the 8-course release scope only for content that is genuinely approved; publish a phased catalogue rather than promise 30 courses. Escalate date or scope choices to the Director.
- **Owner:** Faculty coordinator.

R2 — Production throughput and rework

- **Risk reduction:** use consistent recording specifications, file naming and review checklists; reserve 60 editor-hours/week and batch similar work where practical.
- **Monitoring:** effort-hours completed, work in progress, rework hours and first-pass acceptance.
- **Management / contingency:** reforecast duration using measured throughput; reduce or reorder release scope rather than silently assuming overtime.
- **Owner:** Production lead.

R3 — Payment integration correctness

- **Risk reduction:** use provider-signed callbacks, server-side verification, idempotent transaction handling and a sandbox test matrix covering success, failure, timeout and duplicate callback.
- **Monitoring:** daily comparison of platform transaction status against provider references/settlement reports; alert on orphaned or duplicated transactions.

- **Management / contingency:** suspend affected enrolment activation, reconcile source records and contact impacted learners using an approved communication template.
- **Owner:** Payment/IT owner.

4. Separate issue log (events that have already occurred)

Risks are possible future events; issues are known events. The supplied dashboard summary reports **one defect found by a user** and **16 caught internally**, but provides no detailed issue records. Avoid inventing issue descriptions.

Issue ID	Known fact	Severity / current status	Evidence gap / action
I-01	One user-caught defect is included in the reported DRE calculation.	Unknown; current resolution status not supplied.	Obtain actual issue ID, description, discovery date, severity, owner, fix version and retest evidence.
I-02-I-17	Sixteen defects are reported as caught internally before release.	Aggregate count only.	Attach the defect log and map each record to detection phase and closure status.

5. Lessons learned / closure note

What the planning exercise shows

1. Content readiness, rather than the stated 10-week software build, governs a full 30-course release under the given production assumptions.
2. A phased launch aligns the technology milestone with what can be supported by approved content; the first 8 courses can be considered at W10, while the 30-course plan estimates W18.
3. Capacity arithmetic is useful only when its assumptions are visible. Faculty calendars, editor availability and rework must be monitored as actual data.
4. Revenue figures are transparent target-based projections, not guaranteed income. Dashboards should distinguish targets, pending transactions and verified payments.
5. Certificate rules are small but consequential; boundary tests at 60%, 70%, 80%, 99% and 100% help expose off-by-one and authorization defects.
6. Architecture separation helps teams change video or payment vendors without rewriting learning rules, but external integrations still need real operational ownership and failure handling.

Closure status

This is a planning and documentation baseline prepared from the supplied case statement and dashboard walkthrough. It is **not a declaration that the platform has passed production acceptance**. Before closure or launch approval, attach actual requirement sign-off, schedule baselines, test execution logs, source-code size measurement, detailed defect records, security review and stakeholder approval of the open decisions in the SRS.